

Computational materials and chemistry scientist developing data-driven and AI-enabled approaches for materials discovery and atomistic simulation, with a focus on porous and inorganic materials. My research aims to bridge theory and experiment by building predictive atomistic models and new computational methodologies that can address complex materials problems beyond the reach of conventional approaches. I combine first-principles and molecular simulation methods, including DFT, MD, and GCMC, with emerging tools such as machine-learning interatomic potentials and large language models. Ultimately, I seek to establish computational frameworks that accelerate the discovery, understanding, and design of functional materials.

SKILLS

Programming language	Python
Simulation tools	VASP, LAMMPS, RASPA
Visualizer	Materials studio, VESTA, VMD, OVITO

TECHNICAL EXPERIENCE

BK21 Postdoctoral research fellow 09.2025 — Present
Research Institute of Advanced Materials (RIAM), Seoul National University (Advisor: Prof. Seungwu Han) Seoul, Republic of Korea

- Benchmark of various machine learning interatomic potentials (MLIPs, e.g. SevenNet, MACE, eSEN, UMA, and etc...) (**Pubs. 17**)
- Acceleration of crystal structure prediction using MLIPs to discover unsolved materials.
- Acceleration of high-fidelity materials discovery using MLIPs.

Visiting researcher 07.2023 — 01.2024
Imperial College London (Advisor: Prof. Aron Walsh) London, United Kingdom

- Identifying design principles of covalent-organic frameworks (COFs) for lithium-ion conduction using transformer-based ML model.

Graduate student researcher 03.2016 — 08.2025
Korea Advanced Institute of Science and Technology (KAIST) (Advisor: Prof. Jihan Kim) Daejeon, Republic of Korea

- Data-driven porous materials design for carbon capture in humid conditions and validating them from atomistic level to process level (**Pubs. 16**).
- Development and application of MLIPs for accurate and accelerated modeling of MOF-CO₂/H₂O interactions in direct air capture (**Pubs. 12, 14**).
- Porous materials database construction and high-throughput screening to find candidates for gas fuel storage (**Pubs. 10**)
- ML aided porous materials screening and discovery for gas separations (**Pubs. 1, 2**)
- Support experimental results (especially for chemiresistive sensor) using atomistic scale simulations (**Pubs. 3, 4, 7, 8, 9, 15**).

Intern 12.2018 — 02.2019
SK Hynix Icheon, Republic of Korea

- Worked at DRAM Photo team

EDUCATION

Ph.D. in Chemical and Biomolecular Engineering, KAIST 03.2022 — 08.2025
Dissertation title: Efficient Computational Modeling of Metal-Organic Frameworks for Greenhouse Gas Reduction (Advisor: Prof. Jihan Kim)

M.S. in Chemical and Biomolecular Engineering, KAIST 03.2020 — 02.2022
Thesis title: Finely Tuned Inverse Design of Metal-Organic Frameworks for Selective Xenon Adsorption (Advisor: Prof. Jihan Kim)

B.S. in Chemical and Biomolecular Engineering, KAIST 03.2016 — 02.2020

AWARDS AND HONORS

College of Engineering PhD Dissertation Award (Dissertation Award), College of Engineering @ KAIST	02.2026
Postdoctoral Researcher Fellowship, BK21 Plus Program	09.2025 — 02.2027
International Research Fellowship, BK21 Plus Program	07.2023 — 01.2024
Department Honors Scholarship, KAIST	2018 Spring

PUBLICATIONS

(*: Corresponding author, ‡: Co-first author / 1st author: 7, Co-author: 12)

- [19] Y.Lim‡, H. Jeon‡, J. Kim, J. Kim, and S. Han*, "Agentic programs: an emerging form of scientific software in computational materials science", *arxiv*, 2026 (Perspective)

- [18] J. Kim†, D. Kim†, S. Kim, Y. Lim, and J. Kim*, “Framework-Templated Gas Lattices in Metal-Organic Frameworks”, *Nat. Commun.*, 17, 7873, 2026
- [17] J. Kim†, J. You†, Y. Park, Y. Lim, Y. Kang, J. Kim, H. Jeon, D. Hong, S.Y. Lee, S. Choi, Y. Kim, J.W. Lee, and S. Han*, “Optimizing Cross-Domain Transfer for Universal Machine Learning Interatomic Potentials”, *Nat. Commun.*, 17, 3432, 2026
- [16] Y. Lim, W. Jeong, H. Jung, H.L. Choi, T.-H. Bae, J.H. Lee*, and J. Kim*, “Tuning Pore Chemistry of Metal-Organic Frameworks using Anionic Pillars for Humid-tolerant CO₂ Capture”, *Chem. Mater.*, 37 (21), 8797-8808, 2025
- [15] K. Lee†, M. Kim†, Y. Kwon†, S. Park†, Y. Lim, D. Kwak, J. Jeong, B. Kim, J. Ahn, J. Kim, Y.-H. Cho, I.-D. Kim* and I. Park*, “Enabling Weather-Independent Gas Detection through Deep Learning on Light-Activated Sensors”, *ACS Nano*, 19 (44), 38327-38339, 2025
- [14] Y. Chiang, T. Kreiman, C. Zhang, M.C. Kuner, E.J. Weaver, I. Amin, H. Park, Y. Lim, J. Kim, D. Chrzan, A. Walsh, S.M. Blau, A.S. Krishnapriyan, and M. Asta, “MLIP Arena: Advancing Fairness and Transparency in Machine Learning Interatomic Potentials via an Open, Accessible Benchmark Platform”, *NeurIPS 2025 Datasets and Benchmarks track (spotlight)*, 2025
- [13] D. Kim, Y. Lim, and J. Kim*, “Refining a Generic Force Field for Predicting Phase Transitions in Wine-Rack Metal-Organic Frameworks”, *J. Chem. Theory Comput.*, 21 (11), 5599-5607, 2025
- [12] Y. Lim†, H. Park†, A. Walsh*, and J. Kim*, “Accelerating CO₂ direct air capture screening for metal-organic frameworks with a transferable machine learning force field”, *Matter*, 8 (7), 102203, 2025
- [11] J. Park, H. Kim, Y. Kang, Y. Lim, and J. Kim*, “From Data to Discovery: Recent Trends of Machine Learning in Metal-Organic Frameworks”, *JACS Au*, 4 (10), 3727-4743, 2024 (Perspective)
- [10] Y. Lim†, B. Kim†, and J. Kim*, “Data-Driven Design of Flexible Metal-Organic Frameworks for Gas Storage”, *Chem. Mater.*, 36 (11), 5465-5473, 2024
- [9] S. Park†, M. Kim†, Y. Lim, D.H. Oh, J. Ahn, C. Park, S. Woo, W.C. Jung, J. Kim, and I.D. Kim*, “Dual-photosensitizer synergy empowers ambient light photoactivation of indium oxide for high-performance NO₂ sensing”, *Adv. Mater.*, 36 (24), 2313731, 2024
- [8] J. Shin†, G. Lee†, M. Choi†, H. Jang, Y. Lim, G.S. Kim, S.H. Nam, S.H. Baek, H.C. Song, J. Kim, C.Y. Kang, J.O. Lee*, S. Jeon*, D. Cho*, and J.S. Jang*, “Atomically mixed catalysts on a 3D thin-shell TiO₂ for dual-modal chemical detection and neutralization.”, *J. Mater. Chem. A*, 11 (34), 18195-18206, 2023
- [7] J. Ahn†, S. Park†, D.H. Oh, Y. Lim, J.S. Nam, J. Kim, W.C. Jung, and I.D. Kim*, “Rapid Joule Heating Synthesis of Oxide-Socketed High-Entropy Alloy Nanoparticles as CO₂ Conversion Catalysts.”, *ACS Nano*, 17 (13), 12188-12199, 2023
- [6] G.S. Kim†, Y. Lim†, J. Shin, J. Yim, S. Hur, H.C. Song, S.H. Baek, S.K. Kim, J. Kim*, C.Y. Kang*, and J.S. Jang*, “Breathable MOFs layer on atomically grown 2D SnS₂ for stable and selective surface activation”, *Adv. Sci.*, 10 (17), 2301002, 2023 (Back cover featured)
- [5] J. Park, Y. Lim, S. Lee, and J. Kim*, “Computational Design of Metal-Organic Frameworks with Unprecedented High Hydrogen Working Capacity and High Synthesizability”, *Chem. Mater.*, 35 (1), 9-16, 2023.
- [4] S. Park, Y. Lim, D.H. Oh, J. Ahn, C. Park, M. Kim, W.C. Jung, J. Kim, and I.D. Kim*, “Steering selectivity in detection of exhaled biomarkers over oxide nanofibers dispersed with noble metals”, *J. Mater. Chem. A*, 11 (7), 3535-3545, 2023.
- [3] J.S. Jang*, Y. Lim, H. Shin, J. Kim, and T.G. Yun*, “Bi-directional water-stream behavior on multifunctional membrane for simultaneous energy generation and water purification”, *Adv. Mater.*, 35 (7), 2209076, 2023.
- [2] Y. Lim, and J. Kim*, “Application of transfer learning to predict diffusion properties in metal-organic frameworks”, *Mol. Syst. Des. Eng.*, 7 (9), 1056-1064, 2022.
- [1] Y. Lim, J. Park, S. Lee, and J. Kim*, “Finely tuned inverse design of metal-organic frameworks with user-desired Xe/Kr selectivity”, *J. Mater. Chem. A*, 9 (37), 21175-21183, 2021.

CONFERENCES AND PRESENTATIONS

- [5] “Data-driven discovery of metal-organic frameworks for selective CO₂ capture under humid condition using machine learning interatomic potential”, *ACS Spring 2026*, Atlanta, Georgia, United States (Oral presentation)
- [4] “Enhancing the Screening of Metal-Organic Frameworks for CO₂ Direct Air Capture using Machine Learning Force Field”, *The 26th Asian Workshop on First-Principles Electronic Structure Calculations*, Tsukuba, Japan (Poster presentation)
- [3] “Improving the screening of metal-organic frameworks for CO₂ direct air capture using machine learning force field”, *2025 KICChE Spring Meeting*, Daegu, Republic of Korea (Oral presentation)
- [2] “Data-Driven Design of Adsorption Responsive Flexible Metal-Organic Frameworks”, *The 12th International Conference and Exhibition on Chemistry in Industry*, Dhahran, Kingdom of Saudi Arabia (Poster presentation)
- [1] “Leveraging Knowledge from Adsorption Property to Predict Diffusion Property in Metal-Organic Frameworks”, *2023 KICChE Spring Meeting*, Pusan, Republic of Korea (Poster presentation)